EXPERIMENT 7- CLUSTERING USING K-MEANS AND GMM

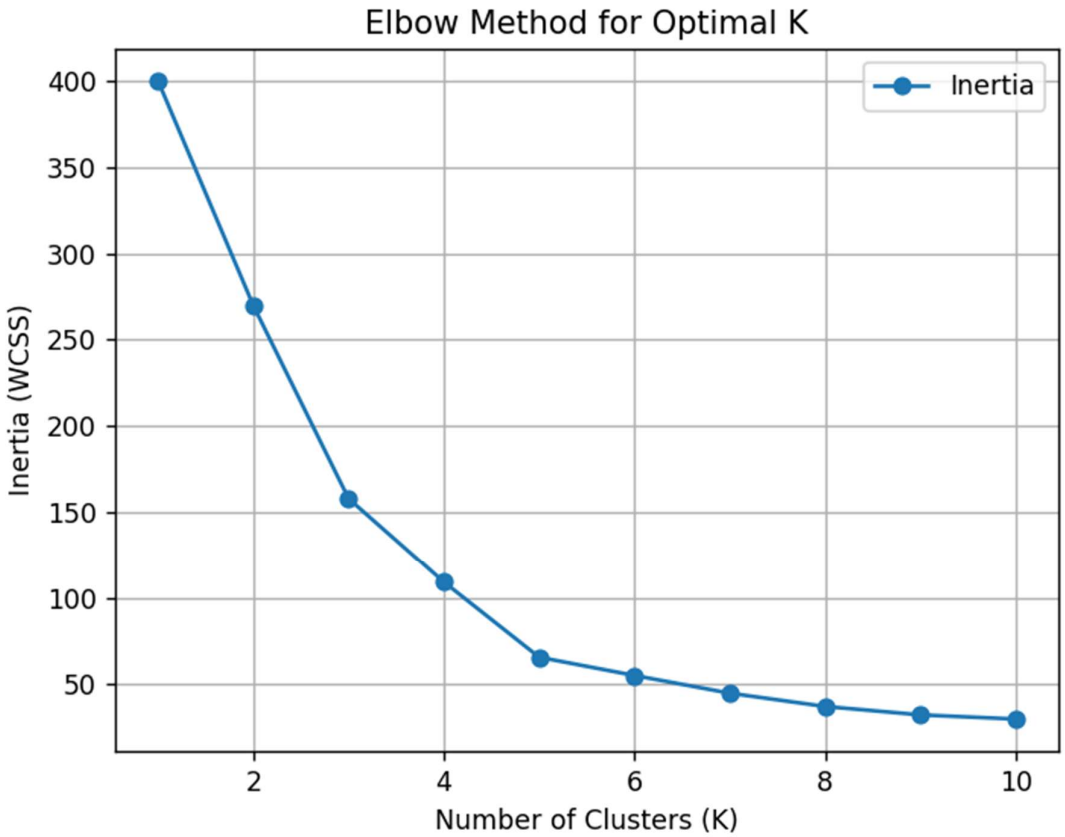
1.SCENARIO 1: K MEANS CLUSTERING

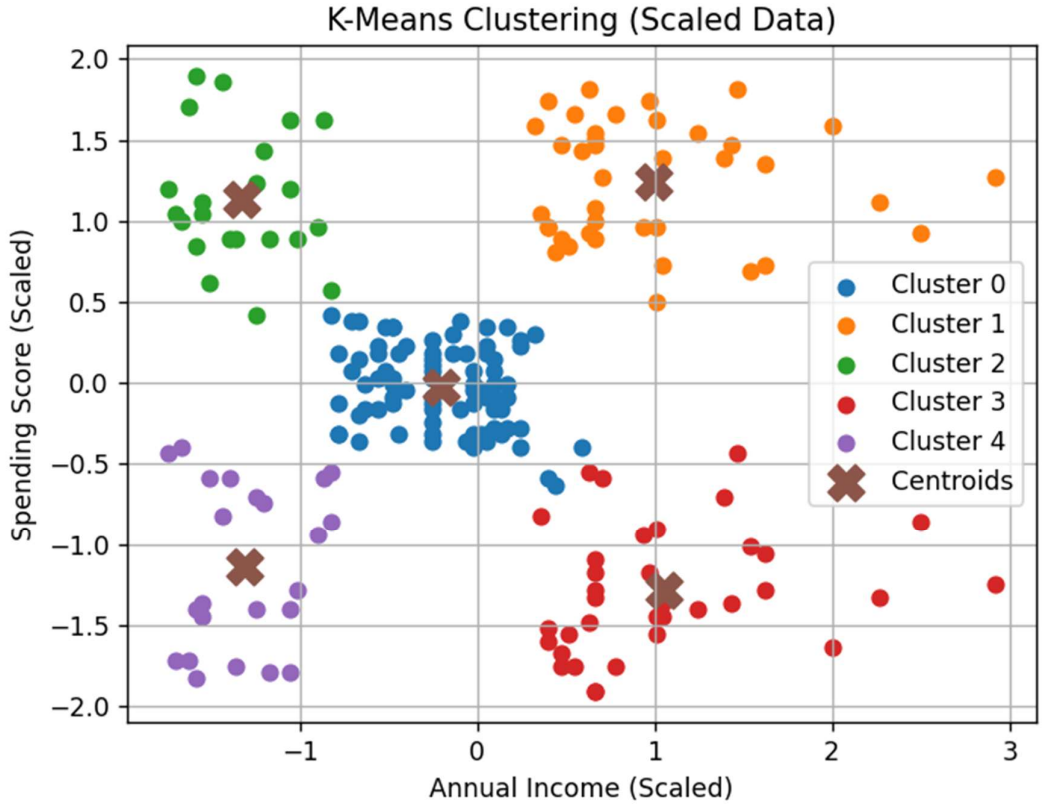
OUTPUT:

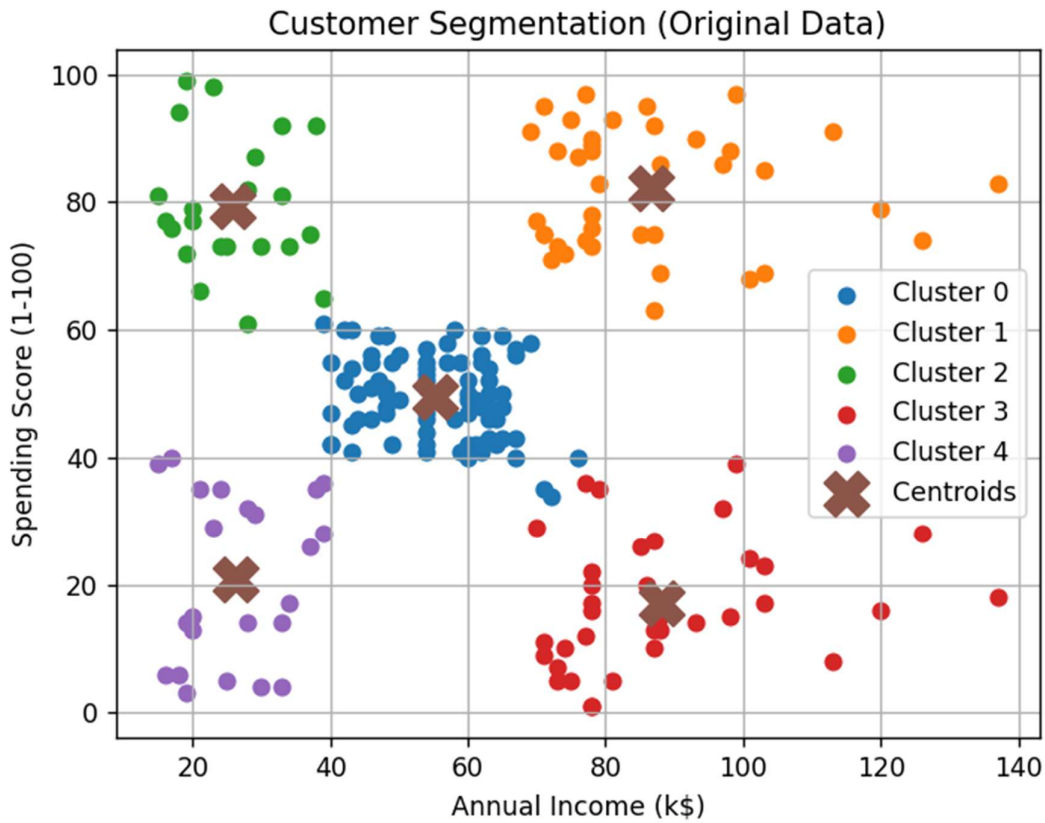
```
JANASREE 24BAD040
First 5 Rows:
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
```

```
Missing Values:
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64

Silhouette Score: 0.5546571631111091

Cluster Centroids:
   Annual Income (k$)  Spending Score (1-100)
0           55.296296              49.518519
1           86.538462              82.128205
2           25.727273              79.363636
3           88.200000              17.114286
4           26.304348              20.913043

Cluster-wise Mean Values:
         Annual Income (k$)  Spending Score (1-100)
Cluster
0                 55.296296              49.518519
1                 86.538462              82.128205
2                 25.727273              79.363636
3                 88.200000              17.114286
4                 26.304348              20.913043
```

PLOTS:


Elbow Method for Optimal K
Inertia
Inertia (WCSS)
400
350
300
250
200
150
100
50
2
4
6
8
10
Number of Clusters (K)


K-Means Clustering (Scaled Data)
Spending Score (Scaled)
Annual Income (Scaled)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Centroids

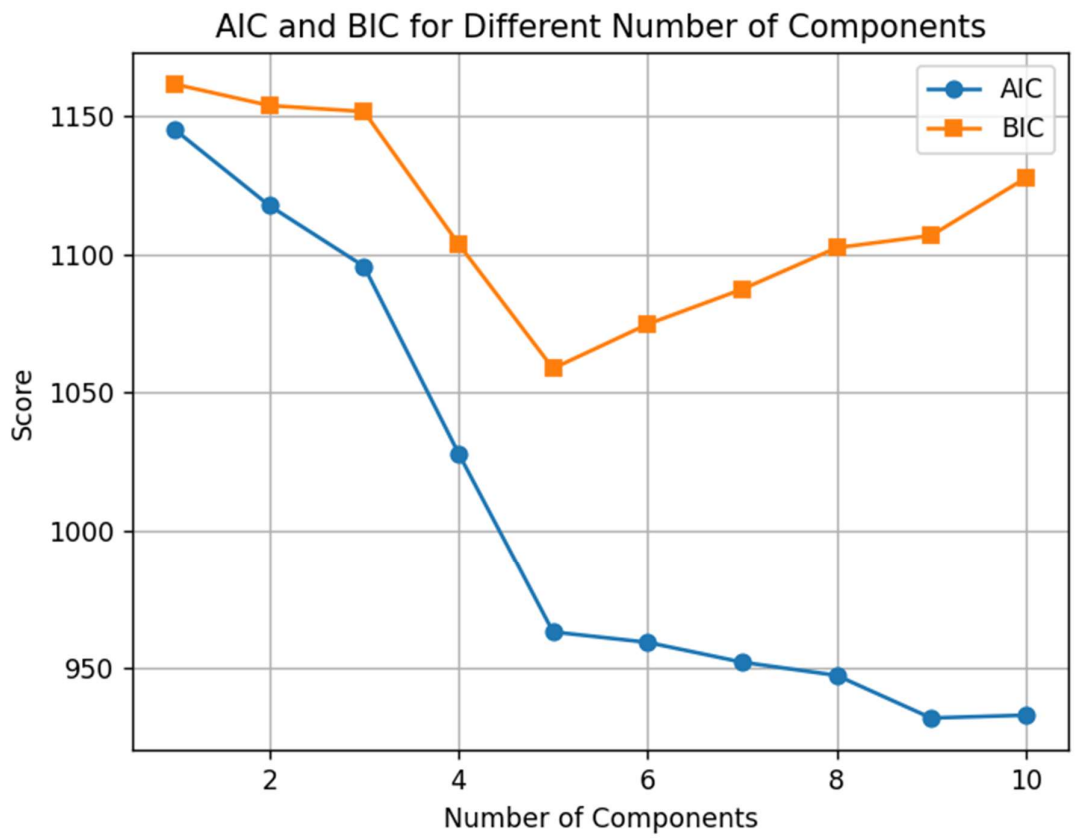


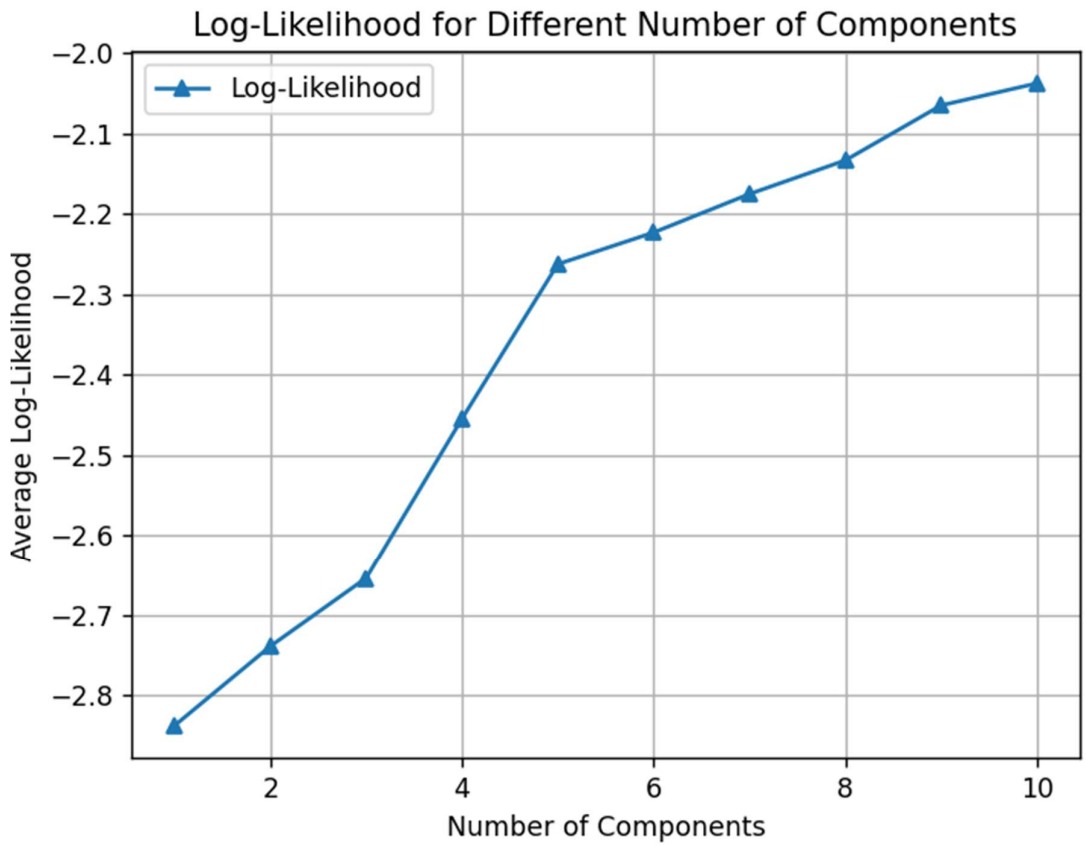
2.SCENARIO 2:GMM

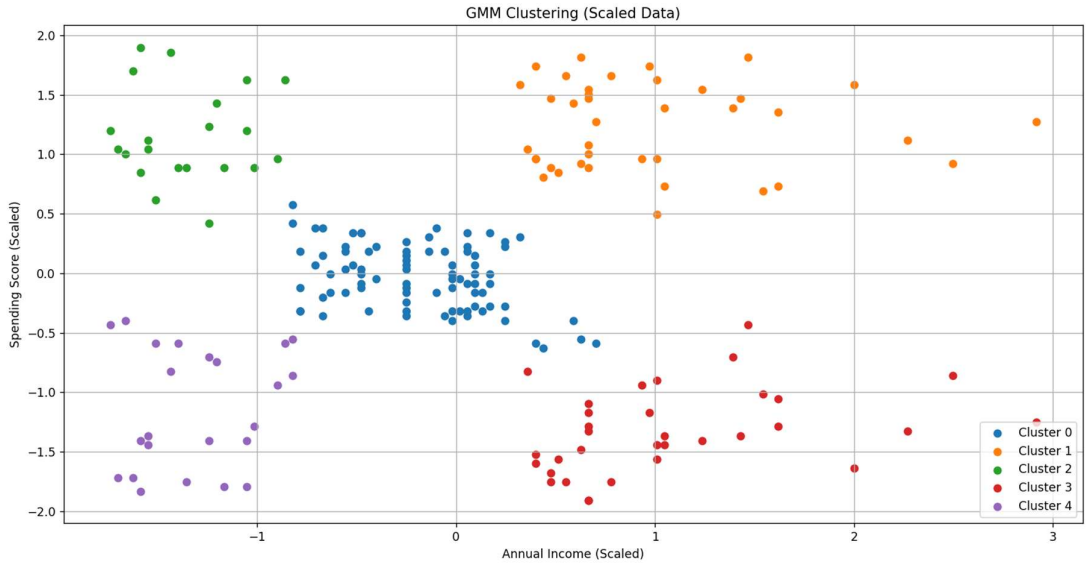
OUTPUT:

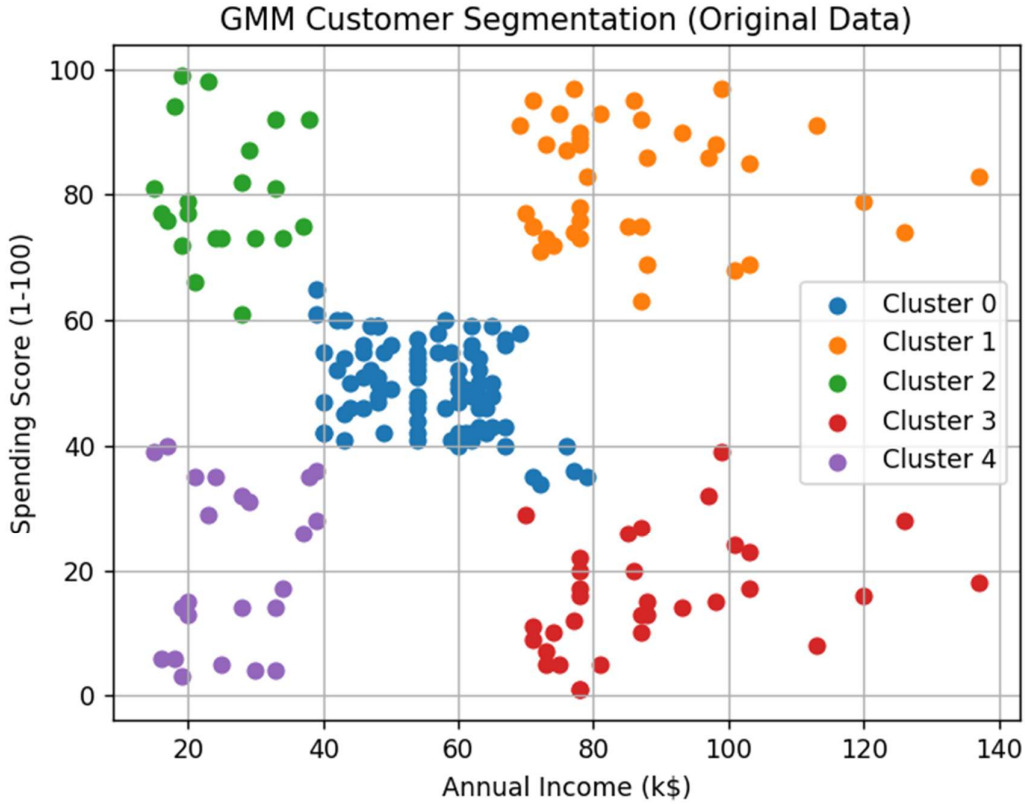
```
======================== RESTART: C:/Users/janas/AppData/Local/Programs
JANASREE 24BAD040
Silhouette Score (GMM): 0.5536892843811245
Log-Likelihood: -2.262502621973241
AIC: 963.0010487892964
BIC: 1058.6522524191894

First 10 Cluster Probabilities:
          0          1          2          3          4
0  1.576897e-05  1.880092e-08  9.520312e-04  5.200131e-08  9.990321e-01
1  1.627762e-05  1.895968e-04  9.997902e-01  4.210134e-23  3.941544e-06
2  2.213394e-19  9.526157e-19  1.225495e-12  1.862233e-04  9.998138e-01
3  1.054448e-04  1.800877e-04  9.996937e-01  3.793901e-21  2.072298e-05
4  6.473014e-05  3.842732e-08  1.556695e-03  4.757227e-08  9.983785e-01
5  1.833287e-04  1.899073e-04  9.995917e-01  1.435839e-20  3.509764e-05
6  9.213688e-19  1.289485e-18  1.408409e-12  2.548375e-04  9.997452e-01
7  1.296572e-08  3.233133e-04  9.996766e-01  8.592922e-29  8.029328e-08
8  4.061493e-20  1.343869e-19  1.906626e-13  4.170864e-04  9.995829e-01
9  1.200053e-03  2.039986e-04  9.983733e-01  1.627723e-18  2.226725e-04

Cluster-wise Mean Values (GMM):
           Annual Income (k$)  Spending Score (1-100)
GMM_Cluster
0                   55.642857               49.369048
1                   86.538462               82.128205
2                   25.095238               80.047619
3                   88.818182               16.000000
4                   26.304348               20.913043
```

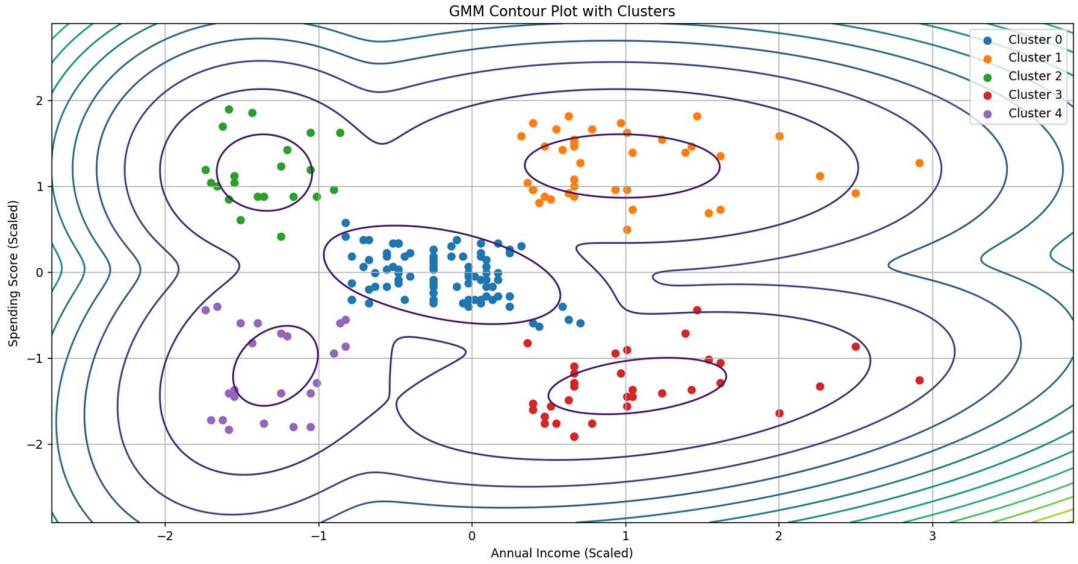
PLOT:

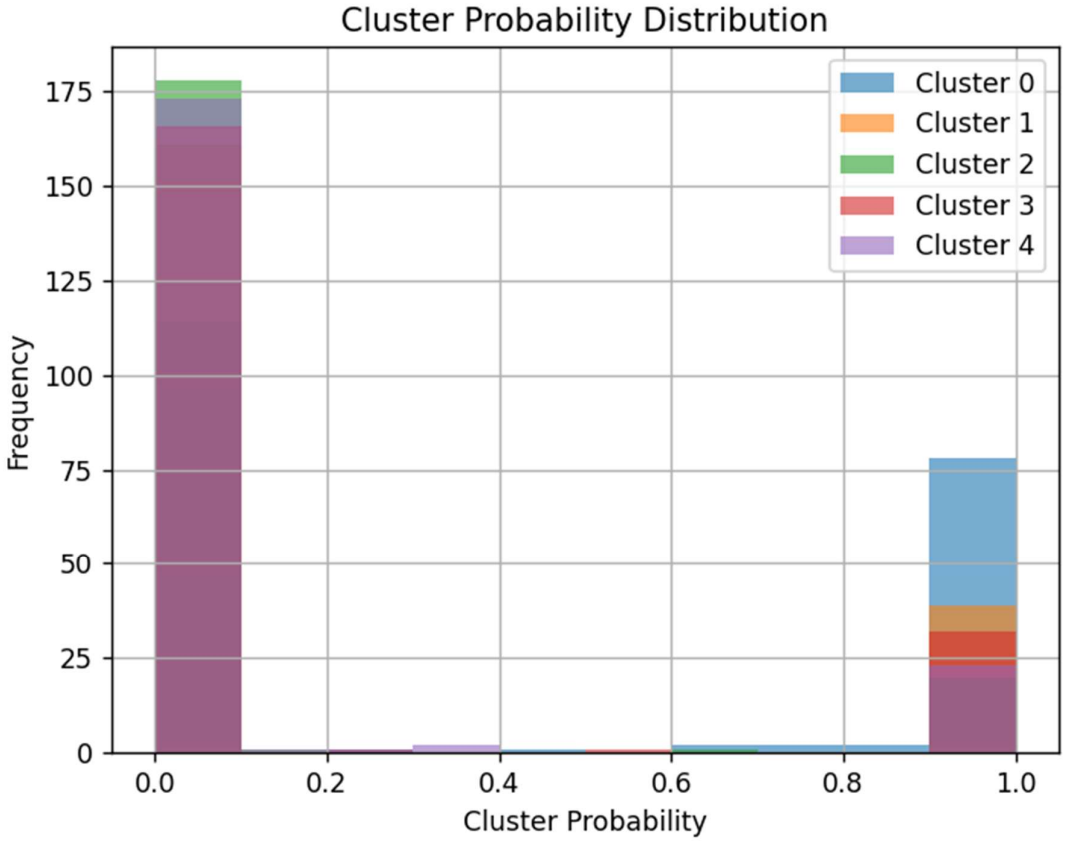

AIC and BIC for Different Number of Components
AIC
BIC
Score
1150
1100
1050
1000
950
2
4
6
8
10
Number of Components


Log-Likelihood for Different Number of Components
Log-Likelihood
Average Log-Likelihood
Number of Components
−2.0
−2.1
−2.2
−2.3
−2.4
−2.5
−2.6
−2.7
−2.8
2
4
6
8
10

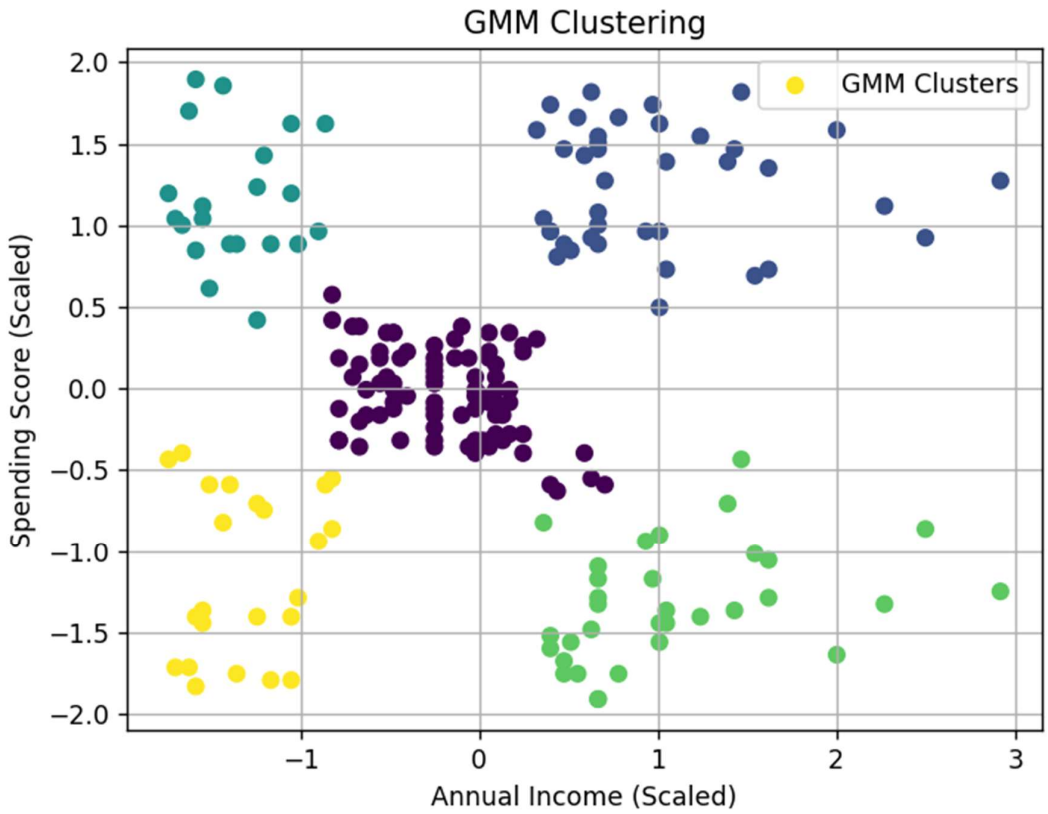

GMM Clustering (Scaled Data)
Spending Score (Scaled)
Annual Income (Scaled)
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4


GMM Customer Segmentation (Original Data)
Spending Score (1-100)
Annual Income (k$)
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4


GMM Contour Plot with Clusters
Spending Score (Scaled)
Annual Income (Scaled)
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4


Cluster Probability Distribution
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Frequency
0
25
50
75
100
125
150
175
0.0
0.2
0.4
0.6
0.8
1.0
Cluster Probability

K-Means Clustering
Spending Score (Scaled)
Annual Income (Scaled)
K-Means Clusters
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3


GMM Clustering
GMM Clusters
Spending Score (Scaled)
Annual Income (Scaled)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3